VijayLaxmi Verma

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SUMMARY

Software Engineer with 3+ years of experience building autonomous motion planning systems for production agricultural machinery, alongside full-stack systems development in C++. Deep applied expertise in path search (A*, Hybrid A*), cost-map based planning, path smoothing, real-time obstacle avoidance, dynamic replanning, and multi-vehicle coordination deployed on real fielded robots. Currently building reliable, deterministic LLM-based agents, with a focus on structured instruction design, evaluation, and consistent output behavior. Solved 550+ algorithmic problems on LeetCode. Seeking to deepen theoretical grounding in motion planning and autonomous decision-making through graduate study in Robotics.

AREAS OF EXPERTISE

Motion Planning & Robotics: A*, Hybrid A*, Cost-Map-Based Search, Path Smoothing, Trajectory/Turn Generation, Obstacle Avoidance, Dynamic Replanning, Multi-Vehicle Coordination, Non-Holonomic Motion Constraints, Embedded Systems (C++)

Languages & Systems: C++, Java, Python, TypeScript, Rust, React.js

Software Engineering: Data Structures & Algorithms, Object-Oriented Design, Design Patterns, SOLID, Performance Optimization, CI/CD, Jenkins, Docker, Kubernetes

Data & Infra: MySQL, MongoDB, RDBMS, SQL/NoSQL, Apache Kafka, AWS, Cloud Architecture

EXPERIENCE

Senior Software Engineer — June 2026 – Current
Arcesium — *Bangalore, India*

- Design, build, and iterate on **LLM-based agents in Python** that produce deterministic, heuristic-driven administrative outputs, prioritizing reliability and precision over open-ended generation.
- Own the full agent development lifecycle — architecture, structured instruction/skill design, testing, and refinement — defining how agents reason through a task and how final answers are presented to end users.
- Build evaluation and testing frameworks to validate agent correctness, consistency, and determinism across edge cases prior to deployment.

Software Engineer (SE) — July 2023 – May 2026
John Deere — *Pune, India*

- Designed and implemented the full **motion planning pipeline in C++** on embedded systems for autonomous agricultural vehicles operating in unstructured field environments — including A*/Hybrid A* search, cost-map construction, path smoothing, and turn/trajectory generation under non-holonomic motion constraints.
- Built real-time obstacle avoidance and dynamic replanning that responds to moving obstacles and changing field conditions, improving operational reliability by **25%**.
- Developed multi-vehicle coordination logic enabling safe, conflict-free operation of multiple autonomous machines working the same field simultaneously.
- Led development of a Web App using React.js, Java, and microservices for fleet/operations management, saving **$2.5M annually** in licensing costs; mentored junior developers and led a cross-functional team to deliver production-ready architecture.
- Conceived and developed a patented innovation enhancing product efficiency, taking the idea from concept to a deployed, measurable solution.

SDE Intern — May 2022 – July 2022
SSM Infotech — *Surat, India*

- Developed a React Native mobile application for Android and iOS to manage employee attendance, including location-based punch-in restricted to predefined areas and automated working-hour calculation.
- Designed advanced SQL queries (multi-table joins, nested subqueries, window functions) for real-time attendance analytics, and built the responsive React.js UI with secure, persistent authentication.

PROJECTS

AI-Assisted Code Refactoring Platform — Nov 2023 – Feb 2024
Python, React.js, GitHub Integration

- Designed a web-based tool that automatically improves code quality, readability, and maintainability using AI models and SOLID/DRY/Clean Code practices, improving developer productivity by 30% and reducing technical debt.

Stock Market Prediction — July 2022 – April 2023
Python, TensorFlow, Keras

- Built a stock market prediction platform using CNNs in Python with chart-based data pipelines, trained in TensorFlow/Keras, and deployed a Streamlit dashboard for real-time predictions and visualization.

EDUCATION

Bachelor of Technology, Computer Science and Engineering — July 2019 – May 2023
Sardar Vallabhbhai National Institute of Technology, Surat — *CGPA: 7.84/10*